Log Normal Distribution

1 Definition

A positive continuous random variable X is said to have a Log-normal distribution if $\log_e X \sim$ a Normal distribution.

Derivation of the PDF:

Let, $\log_e X \sim N\left(\mu, \sigma^2\right)$

The CDF of X is $F_X(x) = P(X \leqslant x) = P\left(\log_e X \leqslant \log x\right) \quad [\because x \text{ is positive }]$
$= P\left(\log_e X \leqslant \log x\right)$ if $0 < x < \infty$

As X is a positive RV, $P(X > 0) = 1 \Rightarrow P(X \leqslant 0) = 0$

The PDF of X is,

$$f_X(x) = \begin{cases} 0, & \text{if } x \leqslant 0 \\ P\left[\dfrac{\log_e X - \mu}{\sigma} \leqslant \frac{\log_e x - \mu}{\sigma}\right] & \text{if } x > 0 \end{cases}$$

$$= \begin{cases} \phi\left(\frac{\log_e x - \mu}{\sigma}\right) \frac{1}{\sigma} \frac{1}{x} & \text{if } 0 < x < \infty \\ 0 & \text{, otherwise} \end{cases}$$

$$= \begin{cases} \frac{1}{x\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{\log_e x - \mu}{\sigma}\right)^2} & , 0 < x < \infty \\ 0 & \text{, otherwise} \end{cases}$$

Notation:

$X \sim \Lambda\left(\mu, \sigma^2\right)$

2 Moments

$$\mu_r' = E\left(X^r\right) = E\left(e^{rY}\right) \text{ where } Y = \log_e X \sim N\left(\mu, \sigma^2\right)$$

$$= \int_{-\infty}^{\infty} e^{ry} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{y-\mu}{\sigma}\right)^2} dy \qquad \begin{array}{l} \frac{y-\mu}{\sigma} = z \\ dy = \sigma dz. \\ y = \sigma z + \mu \end{array}$$

$$= \int_{-\infty}^{\infty} e^{r(\sigma z + \mu)} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}z^2} dz.$$

$$= e^{r\mu} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{r\sigma z} e^{-\frac{1}{2}z^2} dz$$

$$= \frac{e^{r\mu}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}\left[\left(z^2 - 2zr\sigma + r^2\sigma^2\right) - r^2\sigma^2\right]} dz$$

$$= \frac{e^{r\mu}}{\sqrt{2\pi}} e^{\frac{r^2\sigma^2}{2}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(z - r\sigma)^2} dz$$

$$= e^{r\mu+\frac{r^2\sigma^2}{2}} \int_{-\infty}^{\infty} \frac{e^{-\frac{1}{2}(z-r\sigma)^2}}{\sqrt{2\pi}} dz$$

$$= e^{r\mu+\frac{r^2\sigma^2}{2}} \cdot 1 \quad [$$

As the integrand is a PDF of $N(r\sigma, 1)$].

$$\text{Mean } = \mu_1' = e^{\mu+\frac{1}{2}\sigma^2}$$

Variance: $= \mu_2' - {\mu_1}'^2 = e^{2\mu+\sigma^2\left(e^{\sigma^2}-1\right)}$
Harmonic Mean:

$$\frac{1}{H} = E\left(\frac{1}{X}\right) = \mu_{-1}' = \frac{1}{e^{-\mu+\frac{1}{2}\sigma^2}} = e^{\mu-\frac{1}{2}\sigma^2}$$

Geometric Mean:

$$\begin{aligned} & \log G = E\left[\log_e X\right] \quad X \sim \Lambda\left(u, \sigma^2\right) \\ \Rightarrow \quad & \log G = \mu \\ \Rightarrow \quad & G = e^{\mu} \end{aligned}$$

Note that for a log-normal distribution,

$$e^{\mu+\frac{1}{2}\sigma^2} > e^{\mu} > e^{\mu-\frac{1}{2}\sigma^2}$$
$$\Rightarrow AM > GM > HM.$$

Median: If $X \sim \Lambda\left(\mu, \sigma^2\right)$, $\log_e X \sim N\left(\mu, \sigma^2\right)$.
We know, $me\left(\log_e x\right) = \mu \Rightarrow \log_e(me) = \mu$.
If $h(x)$ is monotonic function, $me(h(x)) = h(me(x))$

$$\Rightarrow me(x) = e^{\mu} = GM.$$

3 Mode

The mode of a RV is that value of X for which the PDF For $0 < x < \infty, f_X(x) = \frac{1}{\sigma x\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{\log_e x-\mu}{\sigma}\right)^2}$

$$\Rightarrow \ln f_X(x) = \text{ constant } - \ln x - \frac{1}{2}\left(\frac{\ln x-\mu}{\sigma}\right)^2$$

$$\begin{aligned} \tfrac{d}{dx}\ln f_X(x) &= -\tfrac{1}{x} - \tfrac{1}{2\sigma^2} 2(\ln x-\mu)\tfrac{1}{x} \\ &= -\tfrac{1}{x}\left\{1 + \tfrac{\ln x-\mu}{\sigma^2}\right\} \\ &= \begin{cases} > 0 & x < e^{\mu-\sigma^2} \\ > 0 & x > e^{\mu-\sigma^2} \end{cases} \\ \Rightarrow x &= e^{\mu-\sigma^2} \end{aligned}$$

$f_X(x)$ first increases then attains its maxima at $x = e^{\mu-\sigma^2}$ and finally decreases. Hence, $x = e^{\mu-\sigma^2}$ is the mode.

4 Skewness

$$X \sim \Lambda\left(\mu, \sigma^2\right)$$
$$e^{\mu+\frac{1}{2}\sigma^2} > e^{\mu} > e^{\mu-\sigma^2}$$
$$\Rightarrow \text{ mean } > \text{ median } > \text{ mode}$$

Clearly, log-normal distribution is positively skewed.

5 Quartiles

The pth quartile (ξ_p) of $X \sim N\left(\mu, \sigma^2\right)$ is given by $p = p\left[X \leqslant \xi_p\right]$

$$\begin{aligned} p &= p\left[\log_e X \leqslant \log_e \xi_p\right] \\ &= p\left[\frac{\log_e x - \mu}{\sigma} \leqslant \frac{\log_e \xi_p - \mu}{\sigma}\right] = \Phi\left(\frac{\log_e \xi_p - \mu}{\sigma}\right) \end{aligned}$$

Let $\Phi\left(z_p\right) = p$, where z_p is the pth quartile of $N(0,1)$ distribution, z_p is known.

$$\begin{aligned} \Phi\left(\frac{\log_e \xi_p - \mu}{\sigma}\right) &= \Phi\left(z_p\right) \\ \Rightarrow \ln \xi_p &= \mu + \sigma z_p \\ \Rightarrow \xi_p &= e^{m+\sigma z_p} \end{aligned}$$

Clearly, $z_{3/4} = -z_{1/4} = \theta > 0$ (say)

Hence,

$$\begin{aligned} Q_1 &= \xi_{1/4} = e^{\mu - \sigma\theta} \\ Q_2 &= \xi_{1/2} = e^{\mu} \\ Q_3 &= \xi_{3/4} = e^{\mu + \sigma\theta} \end{aligned}$$

Bowley's measure of skewness:

$$\begin{aligned} S_k &= \frac{(Q_3 - Q_2) - (Q_2 - Q_1)}{(Q_3 - Q_2) + (Q_2 - Q_1)} = \frac{e^{\mu+\sigma\theta} - e^{\mu} - \left(e^{\mu} - e^{\mu-\sigma\theta}\right)}{e^{\mu+\sigma\theta} - e^{\mu} + e^{\mu} - e^{\mu-\sigma\theta}} \\ &= \frac{e^{\mu-\sigma\theta}\left\{e^{2\sigma\theta} - 2e^{\sigma\theta} + 1\right\}}{e^{\mu-\sigma\theta}\left\{e^{2\sigma\theta} - 1\right\}} \\ &= \frac{\left(e^{\sigma\theta} - 1\right)^2}{\left(e^{\sigma\theta} - 1\right)\left(e^{\sigma\theta} + 1\right)} = \frac{e^{\sigma\theta} - 1}{e^{\sigma\theta} + 1} > 0 \end{aligned}$$

As, $\sigma\theta > 0 \Rightarrow e^{\sigma\theta} > 1$, skewness > 0 $\quad \Rightarrow$ positively skewed.

6 Problems

1. If $X \sim N(0,1)$, find the mean and variance of $e^{aX}, a > 0$.

$$\begin{aligned} Y = e^{aX} &\Rightarrow \ln Y = aX \\ \operatorname{Var}(\ln Y) &= a^2 \operatorname{Var}(X) = a^2 \\ \ln Y \sim N(0, a^2) &\Rightarrow Y \sim \Lambda(0, a^2) \end{aligned}$$

2. If X is distributed according to Log-Normal distribution with mean 1 and variance 4, find $P\left[-\frac{1}{2} < X < 2\right]$.

$$
\begin{aligned}
P\left[-\frac{1}{2} < X < 2\right] &= F(2) - F\left(-\frac{1}{2}\right) \\
&= P(X \leqslant 2) - P\left(X \leqslant -\frac{1}{2}\right) \\
&= P\left(\log_e X \leqslant \log_e 2\right) - P\left(\log_e X \leqslant -\log_e 2\right) \\
&= P\left(\frac{\log X - 1}{2} \leqslant \frac{\log_e 2 - 1}{2}\right) - P\left(\frac{\log_e X - 1}{2} \leqslant \frac{-\log_e 2 - 1}{2}\right) \\
&= \Phi\left(\frac{\log_e 2 - 1}{2}\right) - \Phi\left(\frac{-\log 2 - 1}{2}\right) \\
&= \Phi(-0.15) - \Phi(0.85) \\
&= \Phi(0.85) - \Phi(0.15) \\
&= 0.25
\end{aligned}
$$

3. $X \sim \Lambda\left(\mu, \sigma^2\right), E(X) = 1, E\left(X^2\right) = 5$. Find μ, σ^2.
Here, $1 = E(X) = e^{\mu + \frac{1}{2}\sigma^2}, 4 + 1^2 = E\left(X^2\right) = e^{2\mu + 2\sigma^2}$

$$
\therefore \quad \left.\begin{array}{l} 2\mu + \sigma^2 = 0 \\ 2\mu + 2\sigma^2 = \log_e 5 \end{array}\right\} \Rightarrow \begin{array}{l} \sigma^2 = \log 5 \\ \mu = -\frac{1}{2}\log 3 \end{array}
$$